

Cálculo II

Lista 2 - Jeann Victor Batista

①. Encontre a primitiva mais geral da função, isto é, a integral indefinida de $f(x)$ com respeito a x : $\int f(x) dx = F(x) + C$, onde $F'(x) = f(x)$

a) $f(x) = 9x^2 + 5x + 2$

$$\int (9x^2 + 5x + 2) dx \rightarrow 9 \int x^2 dx + 5 \int x dx + \int 2 dx$$

$$9 \cdot \frac{x^3}{3} + 5 \cdot \frac{x^2}{2} + 2x \rightarrow 3x^3 + \frac{5}{2}x^2 + 2x + C$$

A primitiva é: $3x^3 + \frac{5}{2}x^2 + 2x + C$.

b) $f(x) = x^{3/4} + 3^x$

$$\int x^{3/4} dx + \int 3^x dx \rightarrow \frac{x^{3/4+1}}{3/4+1} + \frac{3^x}{\ln(3/4) \cdot 4^x} + C$$

A primitiva é $\frac{4}{7} x^{7/4} + \frac{3^x}{\ln(3/4) \cdot 4^x} + C$

c) $f(x) = \frac{2}{x \cdot \sqrt{x}} + \sqrt[3]{x}$

$$\int \left(\frac{2}{x \cdot x^{1/2}} + x^{1/3} \right) dx \rightarrow \int \left(\frac{2}{x^{3/2}} \right) dx + \int (x)^{1/3} dx \rightarrow \int (2x^{-3/2}) dx + \int (x)^{1/3} dx$$

$$2 \cdot \int x^{-3/2} dx + \int x^{1/3} dx \rightarrow 2 \cdot \frac{x^{-3/2+1}}{-3/2+1} + \frac{x^{1/3+1}}{1/3+1} \rightarrow 2 \cdot \frac{x^{-1/2}}{-1/2} + \frac{x^{4/3}}{4/3}$$

$$-4x^{-1/2} + \frac{3x^{4/3}}{4} + C \rightarrow \text{Essa é a primitiva}$$

d) $f(x) = e^{4x} + 3 \cos x$

$$\int (e^{4x} + 3 \cos x) dx \rightarrow \int e^{4x} dx + 3 \int \cos x dx \rightarrow e^{4x} + 3 \cdot \sin x$$

$$e^{4x} + 3 \sin x + C \rightarrow \text{Primitiva}$$

e) $f(x) = 7x^4 - \sec^2 x$

$$\int (7x^4 - \sec^2 x) dx \rightarrow 7 \cdot \int x^4 dx - \int \sec^2(x) dx \rightarrow 7 \cdot \frac{x^5}{5} - \tan(x) + C$$

$$\frac{7}{5} x^5 - \tan(x) + C \rightarrow \text{Primitiva}$$

$$f) f(x) = 3e^x + \frac{1}{4x} - \sin x$$

$$\int (3e^x + \frac{1}{4x} - \sin x) dx \rightarrow \int 3e^x dx + \frac{1}{4} \int \frac{1}{x} dx - \int \sin x dx$$

$$e^x + \frac{1}{4} \cdot \frac{x^2}{2} - (-\cos x) \rightarrow e^x + \frac{1}{8} x^2 + \cos x + C$$

A primitiva é: $e^x + \ln|x| + \cos x + C$.

② Resolva os problemas de valor inicial (PVI) abaixo

a) $\frac{dy}{dx} = 2x - 5, y(2) = 0$

$$\text{EDO} = \frac{dy}{dx} = 2x - 5 \rightarrow \int dy = \int (2x - 5) dx \rightarrow \int dy = \int (2x - 5) dx$$

$$y = 2 \cdot \frac{x^2}{2} - 5x \rightarrow y = x^2 - 5x + C$$

Usando a condição inicial: $y(2) = 0$

$$y(2) = 2^2 - 5 \cdot 2 + C = 0 \rightarrow 0 = 4 - 10 + C \rightarrow C = -6$$

$$y = x^2 - 5x - 6$$

b) $y' = x^{-1}, y(\pi) = 3$

$$\text{EDO} = \frac{dy}{dx} = x^{-1} \rightarrow dy = x^{-1} dx \rightarrow \int dy = \int x^{-1} dx \rightarrow y = \ln|x| + C$$

Usando a condição inicial: $y = \ln|x| + C$

$$y = \ln|\pi| + C = 3 \rightarrow \ln|\pi| + C = 3 \rightarrow -\ln \pi + 3 = C$$

Portanto $C = -\ln \pi + 3$. // $y = \ln|x| - \ln \pi + 3$.

c) $y' = \frac{1}{x^2} + x, x > 0; y(2) = 1$

$$\text{EDO} = \frac{dy}{dx} = \frac{1}{x^2} + x \rightarrow dy = (\frac{1}{x^2} + x) dx \rightarrow \int dy = \int (\frac{1}{x^2} + x) dx$$

$$y = \int x^{-2} dx + \int x dx \rightarrow y = \frac{x^{-1}}{-1} + \frac{x^2}{2} \rightarrow y = -\frac{1}{x} + \frac{x^2}{2} + C$$

Usando a condição inicial: $y(2) = 1$

$$y = -\frac{1}{2} + \frac{2^2}{2} + C = 1 \rightarrow y = -\frac{1}{2} + 2 + C = 1 \rightarrow y = C = -\frac{1}{2}$$

$$d) \frac{d^2 y}{dx^2} = 2 - 6x, y'(0) = 4, y(0) = 1$$

$$\text{EDO: } d^2 y = (2 - 6x) dx^2 \rightarrow \int d^2 y = \int (2 - 6x) dx^2 \dots$$

$$\int (2 - 6x) dx \rightarrow \int 2 dx - \int 6x dx \rightarrow 2x - 3x^2 \rightarrow (2x - 3x^2 + C)$$

$$C_1 = 2 \cdot 0 - 3 \cdot 0^2 + C = 4 \rightarrow C_1 = C = 4 \rightarrow 2x - 3x^2 + 4$$

$$\text{EDO: } dy (2x - 3x^2 + 4) dx \rightarrow \int dy = \int (2x - 3x^2 + 4)$$

$$y = 2 \int x dx - 3 \int x^2 dx + \int 4 dx \rightarrow y = \frac{2x^2}{2} - \frac{3x^3}{3} + 4x$$

$$y = x^2 - x^3 + 4x + C \rightarrow y = 0^2 - 0^3 + 4 \cdot 0 + C = 1 \rightarrow y = C_2 = 1$$

$$y' = 2x - 3x^2 + 4 \quad C_1 = 4$$

$$y = x^2 - x^3 + 4x + 1 \quad C_2 = 1$$

$$e) y'' = x + \sqrt{x}, y'(1) = 2, y(1) = 1$$

$$\text{EDO: } d^2 y = (x + x^{1/2}) dx^2 \rightarrow \int d^2 y = \int (x + \sqrt{x}) dx^2$$

$$\int x dx + \int x^{1/2} dx \rightarrow \frac{x^2}{2} + \frac{x^{1/2+1}}{1/2+1} \rightarrow \frac{x^2}{2} + \frac{x^{3/2}}{3/2} \rightarrow \frac{x^2}{2} + \frac{2x^{3/2}}{3} + C_1$$

$$C_1 = y'(1) = \frac{1^2}{2} + \frac{2 \cdot 1^{3/2}}{3} + C_1 = 2 \rightarrow y'(1) = \frac{1}{2} + \frac{2}{3} + C_1 = 2 \rightarrow C_1 = 5/6$$

$$\text{EDO: } dy (x^2/2 + 2x^{3/2}/3 + 5/6) dx \rightarrow \int dy (x^2/2 + 2x^{3/2}/3 + 5/6) dx$$

$$y = \frac{1}{2} \int x^2 dx + \frac{2}{3} \int x^{3/2} dx + \int \frac{5}{6} dx \rightarrow y = \frac{1}{2} \cdot \frac{x^3}{3} + \frac{2}{3} \cdot \frac{x^{3/2+1}}{3/2+1} + \frac{5x}{6}$$

$$y = \frac{x^3}{6} + \frac{2}{3} \cdot \frac{x^{5/2}}{5/2} + \frac{5x}{6} \rightarrow y = \frac{x^3}{6} + \frac{4x^{5/2}}{15} + \frac{5x}{6} + C$$

$$y(1) = \frac{5x^3}{30} + \frac{8x^{5/2}}{30} + \frac{25x}{30} + C = 1 \rightarrow y(1) = \frac{5}{30} + \frac{8}{30} + \frac{25}{30} + C = 1$$

$$y(1) = 38/30 + C = 1 \rightarrow C = -8/30$$

$$y = \frac{x^3}{6} + \frac{4x^{5/2}}{15} + \frac{5x}{6} - \frac{8}{30}$$

③ Determine o valor das integrais abaixo usando o teorema da variação total.

a) $\int_{-1}^5 (1+3x) dx$

Primitiva $\int_{-1}^5 (1+3x) dx \rightarrow \int_{-1}^5 1 + \int_{-1}^5 3x \rightarrow x + \frac{3x^2}{2}$

Teorema Variação Total: $f(b) - f(a)$

$$5 + \frac{3 \cdot 5^2}{2} - (-1) + \frac{3 \cdot (-1)^2}{2} \rightarrow 5 + \frac{75}{2} + 1 + \frac{3}{2} = \frac{10 + 75 + 2 + 3}{2} = \frac{89}{2}$$

Portanto o valor da integral é: $44 \frac{1}{2}$.

b) $\int_{-2}^0 (x^2 + x) dx$

Primitiva: $\int_{-2}^0 (x^2 + x) dx \rightarrow \int_{-2}^0 x^2 dx + \int_{-2}^0 x dx \rightarrow (\frac{x^3}{3} + \frac{x^2}{2})$

Teorema Variação Total: $f(b) - f(a)$ $f(b) = 0$.

$$- \left(\frac{-2^3}{3} + \frac{(-2)^2}{2} \right) \rightarrow - \left(\frac{-8}{3} + \frac{4}{2} \right) \rightarrow - \left(\frac{-8}{3} + 2 \right) \rightarrow \frac{2}{3}$$

Portanto o valor da integral é: $2/3$.

c) $\int_0^4 (x^3 - 3x^2 + 8) dx$

Primitiva: $\int_0^4 (x^3 - 3x^2 + 8) dx \rightarrow \int_0^4 x^3 - 3 \int_0^4 x^2 + \int_0^4 8 dx$

$$\frac{x^4}{4} - \frac{3x^3}{3} + 8x \rightarrow \frac{x^4}{4} - x^3 + 8x$$

Teorema Variação Total: $f(b) - f(a) \rightarrow f(a) = 0$

$$\frac{4^4}{4} - 4^3 + 8 \cdot 4 \rightarrow \frac{4}{4} - 1 + 8 \rightarrow \frac{1}{4} - \frac{4}{4} + \frac{32}{4} \rightarrow \frac{29}{4}$$

Portanto a integral é: $29/4$.

d) $\int_{-2}^1 \left(\frac{1}{x^3} + \frac{2}{x} + x \right) dx$

Primitiva: $\int_{-2}^1 (x^{-3} + 2x^{-1} + x) dx \rightarrow \int_{-2}^1 x^{-3} + 2 \int_{-2}^1 x^{-1} + \int_{-2}^1 x$

$$\frac{x^{-2}}{-2} + \ln|x| + \frac{x^2}{2} \rightarrow -\frac{1}{2x^2} + \ln|x| + \frac{x^2}{2}$$

Teorema da Variação total: $f(b) - f(a)$

$$\left(\frac{-1}{2 \cdot 1^2} + 2 \ln |1-1| + \frac{(-1)^2}{2} \right) - \left(\frac{-1}{2 \cdot 2^2} + 2 \ln |1-2| + \frac{(-2)^2}{2} \right)$$

$$\frac{-1}{2} + 0 + \frac{1}{2} - \left(\frac{-1}{8} + 2 \ln 2 + 2 \right) = \frac{1}{8} + 2 \ln 2 + 2$$

$$2 \ln 2 + \frac{1}{8} - \frac{16}{8} \rightarrow -2 \ln 2 - \frac{15}{8}$$

Portanto a integral é: $-2 \ln 2 - \frac{15}{8}$

e) $\int_0^{\pi/2} (x + \cos x) dx$

Primitiva: $\int_0^{\pi/2} (x + \cos x) dx \rightarrow \int_0^{\pi/2} x dx + \int_0^{\pi/2} \cos x dx$

$$\frac{x^2}{2} + \sin x + C$$

Teorema da Variação Total: $f(b) - f(a)$

$$\left(\frac{\pi^2}{2} \right) + \sin \pi - \left(\frac{0^2}{2} + \sin 0 \right) \rightarrow \frac{\pi^2}{2} + 1 - 0 + 0$$

$$\frac{\pi^2}{2} \rightarrow \frac{\pi^2}{2} \cdot \frac{1}{2} \rightarrow \frac{\pi^2}{4} \parallel \frac{\pi^2}{8} + 1$$

Portanto a integral é: $\frac{\pi^2}{8} + 1$

f) $\int_0^{\pi} (2 \sin x - 5^x) dx$

Primitiva: $\int_0^{\pi} (2 \sin x - 5^x) dx \rightarrow \int_0^{\pi} 2 \sin x dx - \int_0^{\pi} 5^x dx$

$$2 \cdot \frac{-\cos x}{\ln 5} - \frac{5^x}{\ln 5} \rightarrow -2 \cos x - \frac{5^x}{\ln 5}$$

Teorema da variação total: $f(b) - f(a)$

$$\left(-2 \cos \pi - \frac{5^{\pi}}{\ln 5} \right) - \left(-2 \cos 0 - \frac{5^0}{\ln 5} \right) \rightarrow \left(-2 \cdot 1 - \frac{5^{\pi}}{\ln 5} \right) - \left(-2 \cdot 1 - \frac{1}{\ln 5} \right)$$

$$\frac{2 \cdot 5^{\pi}}{\ln 5} + \frac{2 \cdot 1}{\ln 5} \rightarrow \frac{4 - 5^{\pi} \cdot 1}{\ln 5}$$

A integral é: $\frac{4 - 5^{\pi} - 1}{\ln 5}$

$$g) \int_0^{\pi/4} \sec^2 x \, dx$$

$$\text{Primitiva: } \int_0^{\pi/4} (\sec^2 x) \, dx \rightarrow \tan x \Big|_0^{\pi/4}$$

$$\text{Teorema da Variação Total: } f(b) - f(a)$$

$$\tan \pi/4 - \tan 0 = 1 - 0 \rightarrow 1$$

A integral é 1.

$$h) \int_0^2 (e^x + x^e) \, dx$$

$$\text{Primitiva: } \int_0^2 (e^x + x^e) \, dx \rightarrow \int_0^2 (e^x) \, dx + \int_0^2 (x^e) \, dx$$

$$\frac{e^x + x^{e+1}}{e+1}$$

$$\text{Teorema da Variação Total: } f(b) - f(a)$$

$$\left(\frac{e^2 + 2^{e+1}}{e+1} \right) - \left(\frac{e^0 + 0^{e+1}}{e+1} \right) \rightarrow \frac{e^2 + 1 + 2^{e+1}}{e+1} \rightarrow \text{Integral}$$

$$i) \int_1^4 \frac{1}{2\sqrt{x}} \, dx$$

$$\text{Primitiva: } \int_1^4 \frac{1}{2\sqrt{x}} \, dx \rightarrow \frac{1}{2} \cdot \int_1^4 \frac{1}{x^{1/2}} \, dx \rightarrow \frac{1}{2} \cdot \int_1^4 x^{-1/2} \, dx \rightarrow \frac{1}{2} \cdot \frac{x^{-1/2+1}}{-1/2+1}$$

$$\frac{1}{2} \cdot \frac{x^{1/2}}{1/2} \rightarrow \frac{1}{2} \cdot \frac{2x^{1/2}}{1} \rightarrow x^{1/2} \rightarrow \sqrt{x}$$

$$\text{Teorema da Variação Total } f(b) - f(a)$$

$$\sqrt{4} - \sqrt{1} \rightarrow 2 - 1 \rightarrow 1$$

1 é a integral

2) Calcule as integrais definidas usando a propriedade $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$

$$a) \int_0^4 f(x) \, dx, \text{ onde } f(x) = \begin{cases} x^2, & \text{se } 0 \leq x \leq 2 \\ 2x, & \text{se } 2 < x \leq 4 \end{cases}$$

$$\int_0^4 f(x) \, dx = \int_0^2 x^2 \, dx + \int_2^4 2x \, dx \rightarrow \frac{x^3}{3} \Big|_0^2 + x^2 \Big|_2^4$$

$$\frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3} \quad \text{e} \quad 4^2 - 2^2 = 12$$

$$\frac{8}{3} + 12 \rightarrow \frac{8}{3} + \frac{36}{3} = \frac{44}{3} \quad \text{que é a integral.}$$

b) $\int_0^3 f(x) dx$, onde $f(x) = \begin{cases} 7-x, & \text{se } x < 2 \\ x+3, & \text{se } x \geq 2 \end{cases}$

$$\int_0^3 f(x) dx = \int_0^2 f(x) dx + \int_2^3 f(x) dx \rightarrow \int_0^2 7-x + \int_2^3 x+3$$

$$\left(7x - \frac{x^2}{2} \Big|_0^2 \right) + \left(\frac{x^2}{2} + 3x \Big|_2^3 \right) \rightarrow$$

$$\left(7 \cdot 2 - \frac{2^2}{2} - 7 \cdot 0 - \frac{0^2}{2} \right) + \left(\frac{3^2}{2} + 3 \cdot 3 - \frac{2^2}{2} - 3 \cdot 2 \right)$$

$$\left(14 - 2 \right) + \left(\frac{9}{2} + 9 - 2 - 6 \right) \rightarrow 12 + \frac{9}{2} + 1 \rightarrow \frac{24}{2} + \frac{9}{2} + \frac{2}{2} = \frac{35}{2}$$

A integral portanto é: $35/2$

c) $\int_{-1}^4 f(x) dx$, onde $f(x) = |x-1|$

$$|x-1| = \begin{cases} 1-x, & \text{se } x < 1, \\ x-1, & \text{se } x \geq 1. \end{cases}$$

$$\int_{-1}^4 f(x) dx = \int_{-1}^1 f(x) dx + \int_1^4 f(x) dx$$

$$\int_{-1}^1 (1-x) dx + \int_1^4 (x-1) dx \rightarrow \left(x - \frac{x^2}{2} \right) \Big|_{-1}^1 + \left(\frac{x^2}{2} - x \right) \Big|_1^4$$

$$\left(1 - \frac{1^2}{2} - (-1) \cdot \frac{(-1)^2}{2} \right) + \left(\frac{4^2}{2} - 4 - \frac{1^2}{2} - 1 \right)$$

$$\left(1 - \frac{1}{2} + 1 - \frac{1}{2} \right) + \left(8 - 4 - \frac{1}{2} - 1 \right)$$

$$\left(13 - \frac{1}{2} \right) + \left(5 - \frac{1}{2} \right) \rightarrow \left(\frac{26}{2} - \frac{1}{2} \right) + \left(\frac{10}{2} - \frac{1}{2} \right)$$

$$\left(\frac{25}{2} + \frac{9}{2} \right) \rightarrow \frac{34}{2} \rightarrow 17$$

A integral portanto é: 17 .

5) Use o teorema fundamental do cálculo para encontrar a derivada da função g -

$$a) G(x) = \int_0^x \sqrt{1 + \sec^2 t} dt$$

$$G(x) = \int_0^x \sqrt{1 + \sec^2 t} dt \rightarrow G(x) = \sqrt{1 + \sec^2 x}$$

A derivada é: $\sqrt{1 + \sec^2 x}$

$$b) G(x) = \int_1^x \ln u du$$

$$G(x) = \int_1^x \ln u du \rightarrow G(x)' = \ln x$$

A derivada é: $\ln x$

$$c) G(x) = \int_x^2 \frac{u^3}{1+u^2} dx$$

$$G(x) = \int_x^2 \frac{u^3}{1+u^2} du \rightarrow - \int_2^x \frac{u^3}{1+u^2} du$$

$$- \left(\frac{x^3}{1+x^2} \right) \rightarrow - \frac{x^3}{1+x^2}$$

A derivada é: $-\frac{x^3}{1+x^2}$

$$d) G(x) = \int_1^{\cos x} t^2 \cdot \sin t dt$$

$$h(x) = \int_1^x (t^2 \cdot \sin t) dt \rightarrow$$

$$h'(x) = (x^2 \cdot \sin x)$$

$$f(x) = \cos x \text{ e } f'(x) = -\sin x$$

$$G(x) = h(g(x)) \rightarrow G'(x) = h'(f(x)) \cdot f'(x)$$

$$G'(x) = (\cos^2 \cdot \sin(\cos)) \cdot -\sin x$$

A derivada portanto é: $(\cos^2 x \cdot \sin(\cos)) \cdot -\sin x$

$$e) G(x) = \int_{-\sqrt{x}}^0 (\tan(t^2)) dt$$

$$h(x) = \int_0^x (\tan(t^2)) dt \quad \begin{cases} f(x) = -\sqrt{x} \\ h'(x) = -\tan(x) \end{cases}$$

$$G(x) = h(f(x)) \quad // \quad G'(x) = h'(f(x)) \cdot f'(x)$$

$$G'(x) = -\tan(\sqrt{x}^2) \cdot \frac{1}{2\sqrt{x}} \rightarrow G'(x) = \frac{-\tan(x)}{2\sqrt{x}}$$

⑥. O valor médio M_f de uma função contínua $f(x)$ no intervalo $[a, b]$ é definido como $M_f = \frac{1}{b-a} \int_a^b f(x) dx$. Calcule o valor médio das funções no intervalo b dado.

a) $f(x) = x^2 - 1, [0, \sqrt{3}]$

$$M_f = \frac{1}{b-a} \cdot \int_a^b f(x) dx$$

$$M_f = \frac{1}{\sqrt{3}-0} \cdot \int_0^{\sqrt{3}} (x^2 - 1) dx$$

Resolvendo integral Definida

$$\text{Primitiva: } \int_0^{\sqrt{3}} (x^2 - 1) dx = \left(\frac{x^3}{3} - x \right) \quad f(a) - f(b)$$

$$\left(\frac{(\sqrt{3})^3}{3} - \sqrt{3} \right) - \left(\frac{0^3}{3} - 0 \right) \rightarrow \frac{\cancel{3} \cdot \sqrt{3}}{\cancel{3}} - \sqrt{3} = 0.$$

Voltando a média:

$$M_f = \frac{1}{\sqrt{3}} \cdot 0 \rightarrow 0 \quad // \quad 0 \text{ valor médio é } 0.$$

b) $f(x) = \ln x, [1, 4]$

$$M_f = \frac{1}{b-a} \cdot \int_a^b (\ln x) dx \rightarrow M_f = \frac{1}{4-1} \cdot \int_1^4 (\ln x) dx$$

$$\text{Primitiva: } \int_1^4 (\ln x) dx \rightarrow \ln|x| \quad f(a) - f(b)$$

$$(\ln|4| - \ln|1|) \rightarrow \ln|4|$$

Voltando a média

$$M_f = \frac{1}{3} \cdot \ln|4| \rightarrow \frac{\ln|4|}{3} = 0 \text{ valor médio}$$

$$c) g(x) = \cos x, [0, \pi/2]$$

$$M_f = \frac{1}{b-a} \cdot \int_0^{\pi/2} \cos x \rightarrow M_f = \frac{1}{\pi/2 - 0}$$

$$\text{Primitiva: } \int_0^{\pi/2} (\cos x) dx \rightarrow +\sin x \quad f(b) - f(a)$$

$$(\sin \pi/2 - (\sin 0)) \rightarrow (-1 + 0) \rightarrow -1$$

$$M_f = \frac{1}{\frac{\pi}{2}} \cdot +1 \rightarrow M_f = \frac{-1}{\frac{\pi}{2}} \rightarrow M_f = \frac{+2}{\pi}$$

$$d) g(x) = \sqrt{x} \quad [1, 4]$$

$$M_f = \frac{1}{b-a} \cdot \int_1^4 \sqrt{x} \rightarrow M_f = \frac{1}{4-1} \cdot \int_1^4 \sqrt{x}$$

$$\text{Primitiva: } \int_1^4 \sqrt{x} \rightarrow \frac{x^{1/2+1}}{1/2+1} \rightarrow \frac{x^{3/2}}{3/2} \rightarrow \frac{2x^{3/2}}{3}$$

$$f(b) - f(a) \rightarrow \frac{2 \cdot 4^{3/2}}{3} - \frac{2 \cdot 1^{3/2}}{3} \rightarrow \frac{2 \cdot 8}{3} - \frac{2}{3} \rightarrow \frac{16 \cdot 2}{3} \rightarrow \frac{14}{3}$$

$$M_f = \frac{1}{3} \cdot \frac{14}{3} \rightarrow M_f = \frac{14}{9}$$

$$e) f(t) = e^{-t} \quad [0, 5]$$

$$M_f = \frac{1}{b-a} \cdot \int_0^5 e^{-t} \rightarrow M_f = \frac{1}{5-0} \cdot \int_0^5 e^{-t}$$

$$\text{Primitiva: } \int_0^5 e^{-t} \rightarrow -e^{-t} \quad f(b) - f(a)$$

$$-e^{-5} - -e^0 \rightarrow -e^{-5} + 1 \rightarrow 1 - \frac{e}{5}$$

$$M_f = \frac{1}{5} \cdot \left(\frac{1}{e^5} - 1 \right) \rightarrow M_f = \frac{1}{5} \cdot \left(\frac{1}{e^5} - 1 \right)$$

$$f) = f(\theta) = \sec \theta \cdot \tan \theta, [0, \pi/4]$$

$$M_f = \frac{1}{\frac{\pi}{4} - 0} \int_0^{\pi/4} \sec \theta \cdot \tan \theta \rightarrow M_f = \frac{4}{\pi} \cdot \int_0^{\pi/4} \sec \theta \cdot \tan \theta$$

$$\text{Primitiva: } \int \sec \theta \cdot \tan \theta dx \rightarrow \sec(x) = f(b) - f(a)$$

$$(\sec \pi/4 - \sec 0) \rightarrow (\sqrt{2} - 1)$$

$$M_f = \frac{1}{\frac{\pi}{4}} \cdot (\sqrt{2} - 1) \rightarrow M_f = \frac{4}{\pi} \cdot (\sqrt{2} - 1)$$

$$M_f = \frac{4\sqrt{2}}{\pi} - \frac{4}{\pi}$$

⑦ Uma partícula move-se ao longo de uma reta com posição $s(t)$ e velocidade instantânea $v(t) = s'(t)$. Seja $a(t) = v'(t) = 2t - 1$ sua aceleração instantânea e $v(0) = -6 \text{ m/s}$ sua velocidade inicial.

a) Encontre $v(t)$ e estude seu sinal para $t \geq 0$.

temos a derivada de $v(t)$, se eu integrar a derivada acho $v(t)$.

$$\int v'(t) = \int (2t - 1) dt \rightarrow 2St - St \rightarrow \frac{2t^2}{2} - t \rightarrow t^2 - t + C$$

$$v(0) = -6 \rightarrow -6 = t^2 - t + C \rightarrow -6 = 0^2 - 0 + C \rightarrow C = -6$$

$$v(t) = t^2 - t - 6$$

Estudo de Sinal: Para $t \geq 0$

$$0 = 0^2 - 0 - 6 = -6$$

$$2 = 2^2 - 2 - 6 = -2$$

$$4 = 4^2 - 4 - 6 = 6$$

$$1 = 1^2 - 1 - 6 = -6$$

$$3 = 3^2 - 3 - 6 = 0$$

Ou seja a sinal de $v(t)$ até o instante 3 é negativo, ou seja, anda no sentido contrário. Já do 3 para frente inverte.

b) Determine o deslocamento da partícula durante o período de tempo $1 \leq t \leq 4$ usando o teorema da variação total, isto é, $\Delta s = s(4) - s(1) = \int_1^4 v(t) dt$, e calcule a velocidade média $v_m = \Delta s / \Delta t$ da partícula nesse período de tempo.

$$\int_1^4 u(t) dt \rightarrow \int_1^4 (t^2 - t - 6) dt \rightarrow \int_1^4 (t^2) dt - \int_1^4 (t) dt - 6 \int_1^4 1 dt$$

$$\left(\frac{t^3}{3} - \frac{t^2}{2} - 6t \right) \Big|_1^4 \quad f(b) - f(a)$$

$$f(b) = \left(\frac{4^3}{3} - \frac{4^2}{2} - 6 \cdot 4 \right) = \left(\frac{64}{3} - \frac{16}{2} - 24 \right) \rightarrow \left(\frac{64}{3} - 8 - 24 \right) = \frac{64 - 32 - 72}{3}$$

$$\frac{64 - 96}{3} = \frac{-32}{3} = f(b)$$

$$f(a) = \left(\frac{1^3}{3} - \frac{1^2}{2} - 6 \cdot 1 \right) = f(a) = \left(\frac{1}{3} - \frac{1}{2} - 6 \right) = \left(\frac{2 - 3 - 36}{6} \right) = \frac{-37}{6}$$

$$f(a) - f(b) = \frac{-32}{3} - \left(\frac{-37}{6} \right) = \frac{-64}{6} + \frac{37}{6} = \frac{-27}{6} = \Delta s$$

$$\frac{\Delta s}{\Delta t} = \frac{-27/6}{4 - 1} \rightarrow \frac{-27}{6 \cdot 3} \rightarrow \frac{-27}{18}$$

A velocidade média no período é = -1,5 m/s

c) Determine a distância percorrida pela partícula nesse mesmo período de tempo, isto é, calcule $\int_1^4 |v(t)| dt$. (O módulo de $v(t)$ é para incluir as distâncias percorridas com velocidade negativa que foram subtraídas no cálculo do deslocamento.)

No intervalo de 1 a 4, de 1 a 3 o sinal é negativo, e de 3 a 4 +.

$$\text{Portanto, } \int_1^4 |v(t)| dt = \int_1^3 -(v(t)) dt + \int_3^4 (v(t)) dt$$

Vamos aos cálculos.

$$\int_1^3 -(v(t)) dt = \int_1^3 (-t^2 + t + 6) dt \rightarrow -\int_1^3 (t^2) dt + \int_1^3 (t) dt + 6 \int_1^3 1 dt$$

$$-\frac{t^3}{3} + \frac{t^2}{2} + 6t \Big|_1^3 \quad f(b) - f(a)$$

$$f(b) = -\frac{3^3}{3} + \frac{3^2}{2} + 6 \cdot 3 \rightarrow \frac{-27}{3} + \frac{9}{2} + 18 \rightarrow \frac{-9}{2} + \frac{9}{2} + 18 \rightarrow \frac{-18 + 9 + 36}{2} = \frac{27}{2}$$

$$f(a) = -\frac{1^3}{3} + \frac{1^2}{2} + 6 \cdot 1 \rightarrow \frac{-1}{3} + \frac{1}{2} + 6 = \frac{-2}{6} + \frac{3}{6} + \frac{36}{6} = \frac{37}{6}$$

$$\frac{27}{2} - \frac{37}{6} = \frac{81}{6} - \frac{37}{6} = \frac{44}{6} = \frac{22}{3} = \int_1^4 |v(t)| dt$$

$$\int_3^4 |V(t)| dt \rightarrow \int_3^4 (t^3 - t - 6) dt \rightarrow \int_3^4 (t^3) dt - \int_3^4 (t) dt - 6 \int_3^4 1 dt$$

$$\left(\frac{t^4}{4} - \frac{t^2}{2} - 6t \right) \rightarrow f(b) - f(a)$$

$$f(b) = \frac{4^4}{4} - \frac{4^2}{2} - 6 \cdot 4 = f(b) = \frac{64}{4} - \frac{16}{2} - 24 \rightarrow f(b) = \frac{64}{4} - 8 - 24 \rightarrow f(b) = \frac{64}{4} - 32$$

$$f(b) = \frac{64}{4} - \frac{96}{4} = f(b) = \frac{-32}{4}$$

$$f(a) = \frac{3^4}{4} - \frac{3^2}{2} - 6 \cdot 3 \rightarrow f(a) = \frac{27}{4} - \frac{9}{2} - 18 \rightarrow f(a) = 9 - \frac{9}{2} - 18$$

$$f(a) = \frac{18}{2} - \frac{9}{2} - \frac{36}{2} = \frac{-27}{2}$$

$$f(b) - f(a) = \left(\frac{-32}{4} - \left(\frac{-27}{2} \right) \right) = \frac{-64}{8} + \frac{81}{8} = \frac{17}{8} = \int_3^4 (-V(t)) dt$$

Portanto:

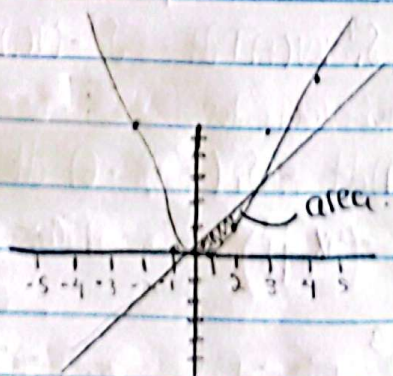
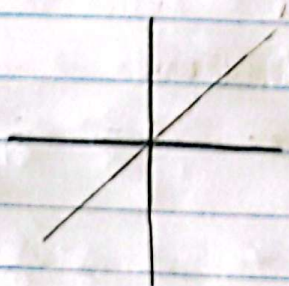
$$\int_3^4 |V(t)| dt = \frac{22}{3} + \frac{17}{6} = \frac{44}{6} + \frac{17}{6} = \frac{61}{6}$$

Então a velocidade é: 10,166...

8) Determine a área da região delimitada por:

a) $y = f(x) = x$ e $y = g(x) = x^2 - x$.

$$\text{Área} = \int_a^b (f(x) - g(x)) dx$$



$$f(x) = G(x) \rightarrow x = x^2 - x \rightarrow x^2 - 2x \rightarrow x(x-2)$$

As raízes portanto são: 2 e 0.

$$A = \int_0^2 [f(x) - G(x)] dx \rightarrow A = \int_0^2 x - (x^2 - x) \rightarrow A = \int_0^2 x - x^2 + x \rightarrow A = \int_0^2 (-x^2 + 2x) dx$$

$$\text{Primitiva: } \int_0^2 (-x^2 + 2x) dx = \int_0^2 (-x^2) dx + 2 \int_0^2 x dx \rightarrow \left(-\frac{x^3}{3} + x^2 \right)$$

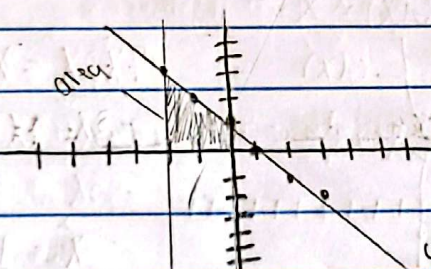
$$f(b) - f(a)$$

$$\rightarrow \left(-\frac{2^3}{3} + 2^2 \right) - \left(-\frac{0^3}{3} + 0^2 \right) = \left(-\frac{8}{3} + 4 \right) - 0 \rightarrow -\frac{8}{3} + \frac{12}{3} = \frac{4}{3}$$

Portanto a área da região delimitada é: 4/3.

b) $y = f(x) = -x + 1$, o eixo x e as retas $x = -2$ e $x = 0$

Gráfico:



$$f(x) = -x + 1$$

$$G(x) = 0$$

$$-x + 1 = 0 \rightarrow x = 1$$

Intervalo entre -2 e 0

$$\int_{-2}^0 (-x + 1) dx \rightarrow \int_{-2}^0 x dx + \int_{-2}^0 1 dx \rightarrow \frac{-x^2}{2} + x$$

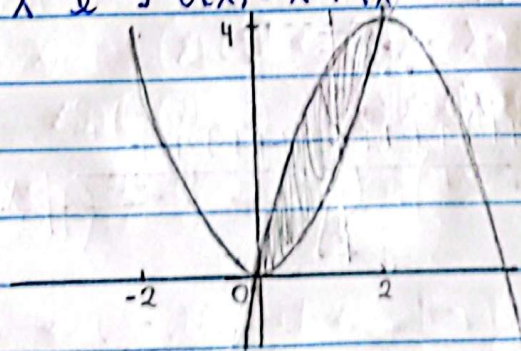
$$f(b) - f(a)$$

$$\left(-\frac{0^2}{2} + 0 \right) - \left(-\frac{(-2)^2}{2} - 2 \right) \rightarrow 0 - \left(-\frac{4}{2} - 2 \right) \rightarrow 0 - (-4) = 4$$

Portanto a área da região delimitada é: 4

c) $y = f(x) = x^2$ e $y = G(x) = -x^2 + 4x$

Gráfico



$$x^2 = -x^2 + 4x$$

$$2x^2 - 4x = 0$$

$$2x(x-2) \rightarrow 2 \cdot 0$$

$$x^2 + 3x - 2$$

$$\int_0^2 (G(x) - f(x)) dx \rightarrow \int_0^2 (-x^2 + 4x - x^2) dx \rightarrow \int_0^2 (-2x^2 + 4x) dx$$

$$\text{Primitiva: } \int_0^2 (-2x^2 + 4x) dx \rightarrow -2 \int_0^2 (x^2) dx + 4 \int_0^2 x dx \rightarrow -\frac{2x^3}{3} + \frac{4x^2}{2}$$

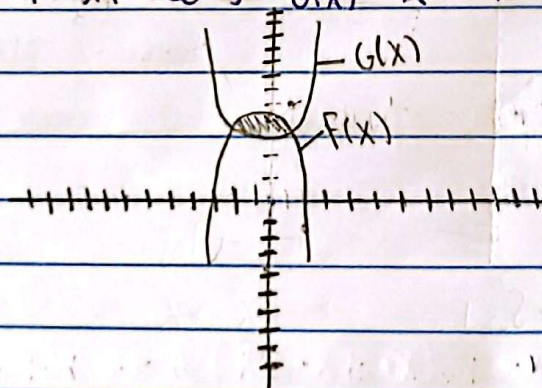
$$-\frac{2x^3}{3} + 2x^2 \rightarrow f(b) - f(a)$$

$$\left(-\frac{2 \cdot 2^3}{3} + 2 \cdot 2^2 \right) - \left(-\frac{2 \cdot 0^3}{3} + 2 \cdot 0^2 \right)$$

$$\frac{-16 + 8}{3} - 0 \rightarrow \frac{-16 + 24}{3} \rightarrow \frac{8}{3}$$

A área da região delimitada é : $\frac{8}{3}$.

$$d) y = f(x) = 7 - 2x^2 \text{ e } y = G(x) = x^2 + 4$$



$$7 - 2x^2 = x^2 + 4$$

$$-3x^2 = -3$$

$$3x^2 = 3$$

$$x^2 = 1$$

$$x = 1 \text{ ou } -1$$

$$\int_{-1}^1 (f(x) - G(x)) dx \rightarrow \int_{-1}^1 (7 - 2x^2 - x^2 - 4) dx \rightarrow \int_{-1}^1 (-3x^2 + 3) dx$$

$$-\frac{3x^3}{3} + 3x \rightarrow -x^3 + 3x$$

$$f(b) - f(a)$$

$$\left(-(-1)^3 + 3 \cdot (-1) \right) - \left(-(1)^3 + 3 \cdot 1 \right) = +2 - (-2) \rightarrow 4$$

A área da região é : 4.

9) Calcule a área assinalada nas figuras a seguir.

a) $f(x) = 4 - x^2$ e $G(x) = -x + 2$

No intervalo de $[-2, -1]$...

$$\int_{-2}^{-1} (f(x) - G(x)) dx \rightarrow \int_{-2}^{-1} (4 - x^2 - (-x + 2)) dx \rightarrow \int_{-2}^{-1} (4 - x^2 + x - 2) dx$$

$$\int_{-2}^{-1} (-x^2 + x + 2) dx \rightarrow -\int_{-2}^{-1} x^2 + \int_{-2}^{-1} x + 2 \int_{-2}^{-1} 1 dx$$

$$\left(-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \quad f(b) - f(a)$$

$$\begin{aligned} & \xrightarrow{f(b)} \\ & -\frac{(-1)^3}{3} + \frac{(-1)^2}{2} + 2 \cdot (-1) \rightarrow \frac{1}{3} + \frac{1}{2} - 2 \rightarrow \frac{2}{6} + \frac{3}{6} - \frac{12}{6} = \frac{-7}{6} \end{aligned}$$

$$\begin{aligned} f(a) &= -\frac{(-2)^3}{3} + \frac{(-2)^2}{2} + 2 \cdot (-2) \rightarrow f(x) = \frac{8}{3} + \frac{4}{2} - 4 \rightarrow f(x) = \frac{8}{3} + 2 - 4 = f(x) = \frac{8}{3} - 2 = \frac{8}{3} - \frac{6}{3} = \frac{2}{3} \\ & \quad 18/3 - 6/3 = 12/3 = 4 \end{aligned}$$

$$-7/6 - 2/3 = -7/6 - 4/6 = -11/6$$

A primeira área é $11/6$.

No intervalo de $[-1, 2]$...

$$\int_{-1}^2 (G(x) - f(x)) dx \rightarrow \int_{-1}^2 ((-x + 2) - (4 - x^2)) dx \rightarrow \int_{-1}^2 (x^2 - x - 2) dx$$

$$\text{Primitiva} = \int_{-1}^2 (x^2) dx - \int_{-1}^2 (x) dx - 2 \int_{-1}^2 1 dx \rightarrow \frac{x^3}{3} - \frac{x^2}{2} - 2x$$

$$f(b) - f(a)$$

$$f(b) = \left(\frac{2^3}{3} - \frac{2^2}{2} - 2 \cdot 2 \right) \rightarrow \frac{8}{3} - \frac{4}{2} - 4 \rightarrow \frac{8}{3} - 2 - 4 \rightarrow \frac{8}{3} - 6 = \frac{8}{3} - \frac{18}{3} = \frac{-10}{3}$$

$$f(a) = \left(\frac{-1^3}{3} - \frac{(-1)^2}{2} - 2 \cdot (-1) \right) \rightarrow \frac{-1}{3} - \frac{1}{2} + 2 \rightarrow \frac{-2}{6} - \frac{3}{6} + \frac{12}{6} = \frac{7}{6}$$

$$\frac{-10}{3} - \frac{7}{6} = \frac{-20}{6} - \frac{7}{6} = \frac{-27}{6} = \left(\frac{-9}{2} \right)$$

No intervalo de $[2, 3]$

$$\int_2^3 (f(x) - G(x)) dx = \left(-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \rightarrow f(b) - f(a)$$

$$f(b) = -\frac{3^3}{3} + \frac{3^2}{2} + 2 \cdot 3 = -\frac{27}{3} + \frac{9}{2} + 6 \rightarrow -9 + 4.5 + 6 \rightarrow -9 + 10.5 = 1.5$$

$$f(a) = -\frac{2^3}{3} + \frac{2^2}{2} + 2 \cdot 2 = -\frac{8}{3} + 2 + 4 = -2.67 + 6 = 3.33$$

$$\frac{3}{2} - \frac{10}{3} \rightarrow \frac{9}{6} - \frac{20}{6} = \frac{-11}{6}$$

Portanto a área total é: $11/6 + 9/2 + 11/6 = 22/6 + 27/6 = 49/6$

b) $f(x) = x^3/3 - x$ e $G(x) = x/3$.

Para calcular os limites dos intervalos:

$$\left(\frac{x^3}{3} - x\right) = \frac{x}{3} \rightarrow \frac{x^3}{3} - \frac{x}{3} - x \rightarrow x \left(\frac{x^2}{3} - \frac{1}{3} - 1\right) \rightarrow x \left(\frac{x^2 - 4}{3}\right)$$

Portanto as raízes são: $(-2, 0, 2)$.

Intervalo de $[-2, 0]$.

$$\int_{-2}^0 (f(x) - G(x)) dx \rightarrow \int_{-2}^0 (x^3/3 - x - x/3) dx \rightarrow \frac{1}{3} \int_{-2}^0 (x^3) dx - \int_{-2}^0 (x) dx - \frac{1}{3} \int_{-2}^0 (x) dx$$

$$\left(\frac{1}{3} \cdot \frac{x^4}{4} - \frac{x^2}{2} - \frac{1}{3} \cdot \frac{x^2}{2}\right) \rightarrow \left(\frac{x^4}{12} - \frac{x^2}{2} - \frac{x^2}{6}\right) \rightarrow \text{Primitiva}$$

$$f(b) - f(a) \quad f(b) = 0$$

$$f(a) = \frac{-2^4}{12} - \frac{(-2)^2}{2} - \frac{(-2)^2}{6} \rightarrow f(a) = \frac{16}{12} - \frac{2}{2} - \frac{2}{6} \rightarrow f(a) = \frac{16}{12} - \frac{24}{12} - \frac{8}{12} = \frac{-16}{12}$$

Intervalo de $[0, 2]$

$$\int_0^2 (G(x) - f(x)) dx \rightarrow \int_0^2 (x/3 - x^3/3 + x) dx \rightarrow \frac{1}{3} \int_0^2 (x) dx - \frac{1}{3} \int_0^2 (x^3) dx + \int_0^2 (x) dx$$

$$\left(\frac{1}{3} \cdot \frac{x^2}{2} - \frac{1}{3} \cdot \frac{x^4}{4} + \frac{x^2}{2}\right) \rightarrow \frac{x^2}{6} - \frac{x^4}{12} + \frac{x^2}{2} \rightarrow \text{Primitiva}$$

$$f(b) - f(a) \quad f(a) = 0$$

$$f(b) = \frac{2^2}{6} - \frac{2^4}{12} + \frac{2^2}{2} \rightarrow f(b) = \frac{4}{6} - \frac{16}{12} + \frac{4}{2} \rightarrow f(b) = \frac{8}{12} - \frac{16}{12} + \frac{24}{12} = f(b) = \frac{16}{12}$$

Intervalo de $[2, 3]$

$$\int_2^3 (f(x) - G(x)) dx \rightarrow \int_2^3 (x^3/3 - x - x/3) dx \rightarrow \frac{1}{3} \int_2^3 (x^3) dx - \int_2^3 (x) dx - \frac{1}{3} \int_2^3 (x) dx$$

Que possui primitiva: $(x^4/12 - x^2/2 - x^2/6)$ $f(b) - f(a)$

$$f(b) = \frac{3^4}{12} - \frac{3^2}{2} - \frac{3^2}{6} \rightarrow \frac{81}{12} - \frac{9}{2} - \frac{9}{6} \rightarrow \frac{81}{12} - \frac{54}{12} - \frac{18}{12} \rightarrow \frac{9}{12} \rightarrow 3/4$$

$$f(a) = \frac{2^4}{12} - \frac{2^2}{2} - \frac{2^2}{6} \rightarrow \frac{16}{12} - \frac{4}{2} - \frac{4}{6} \rightarrow \frac{16}{12} - \frac{24}{12} - \frac{8}{12} \rightarrow -16/12 \rightarrow -4/3$$

$$3/4 - (-4/3) \rightarrow 3/4 + 4/3 \rightarrow \frac{9}{12} + \frac{16}{12} \rightarrow \frac{25}{12}$$

Área total: $4/3 + 16/12 + 25/12 = \frac{16}{12} + \frac{16}{12} + \frac{25}{12} = \frac{57}{12} = \frac{19}{4}$

c) $f(y) = 12y^2 - 12y^3$ e $G(y) = 2y^2 - 2y$

Para calcular os limites dos intervalos:

$$12y^2 - 12y^3 = 2y^2 - 2y \rightarrow -12y^3 + 10y^2 + 2y \rightarrow -6y^3 + 5y^2 + y \rightarrow y(-6y^2 + 5y + 1)$$

Devo achar as raízes das equações:

$$-5 \pm \sqrt{25 - 4 \cdot (-6) \cdot 1} \rightarrow -5 \pm \sqrt{25 + 24} \rightarrow (-5 + 7)/-12 \text{ e } (-5 - 7)/-12$$

Portanto as raízes são: $0, -1/6, 1$.

Como estamos trabalhando com o y e não x : os limites de integração: 0 e 1 .

$$\int_0^1 (f(y) - G(y)) dy \rightarrow \int_0^1 (12y^2 - 12y^3 - 2y^2 + 2y) dy \rightarrow \int_0^1 (-12y^3 + 10y^2 + 2y) dy$$

$$\left(-\frac{12y^4}{4} + \frac{10y^3}{3} + \frac{2y^2}{2} \right) \rightarrow \left(-3y^4 + \frac{10y^3}{3} + y^2 \right) = \text{Primitiva}$$

$f(b) - f(a) \quad f(a) = 0$

$$f(b) = -3 \cdot 1^4 + \frac{10 \cdot 1^3}{3} + 1^2 \rightarrow \left(-3 + \frac{10}{3} + 1 \right) \rightarrow \frac{-9}{3} + \frac{10}{3} + \frac{3}{3} \rightarrow \frac{13 - 9}{3} = \frac{4}{3}$$

Portanto a área assinalada é $4/3$.

d) $y = x, y = 1, y = x^{2/4}$. $f(x) = x$ e $f(x) = x^{2/4}$.

$$x = x^{2/4} \rightarrow x - x^{2/4} = 0 \rightarrow x(1 - x^{1/4}) \quad \text{// raízes são: } 0, 1.$$

Portanto o intervalo seria: $0, 1$ se não houvesse o limitante $y=1$, Logo $[0, 1]$.

$$\int_0^1 (f(x) - G(x)) dx \rightarrow \int_0^1 (x - x^{2/4}) dx \rightarrow \left(\frac{x^2}{2} - \frac{x^{3/2}}{3/2} \right) \text{ é a primitiva}$$

$$f(b) - f(a) \quad f(a) = 0$$

$$f(b) = \left(\frac{2^2}{2} - \frac{2^{3/2}}{3/2} \right) \rightarrow f(b) = \frac{2}{2} - \frac{8}{12} \rightarrow f(b) = 2 - \frac{2}{3} \rightarrow f(b) = \frac{6}{3} - \frac{2}{3} = f(b) = \frac{4}{3}$$

Portanto a área seria: $4/3$.